



Εθνικό Μετσόβιο Πολυτεχνείο
Σχολή Ηλεκτρολόγων Μηχανικών και Μηχανικών Υπολογιστών

Ψηφιακά Συστήματα VLSI

2η Εργαστηριακή Άσκηση
Σχεδιασμός Αριθμητικών Μονάδων με Ιεραρχική Σχεδίαση

ΟΝΟΜΑΤΕΠΩΝΥΜΑ	Ευαγγελία Αγγελάκη (el21037) Μαρία Φατούρου(el21077)
ΕΞΑΜΗΝΟ	8ο
ΟΜΑΔΑ	<u>Βάρδια Β/Ομάδα 17</u>

Στην παρούσα εργαστηριακή άσκηση, κληθήκαμε κυρίως να εξοικειωθούμε με τη μεθοδολογία σχεδίασης με χρήση δομικής περιγραφής (Structural description). Έτσι σχεδιάσαμε τις παρακάτω δομές, για την κάθε μία χρησιμοποιώντας της προηγούμενη με ιεραρχική δομή

1. Half adder-Dataflow

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
```

```
entity HA is
  Port ( in1 : in STD_LOGIC;
        in2 : in STD_LOGIC;
        s : out STD_LOGIC;
        c : out STD_LOGIC);
end HA;
```

architecture Dataflow of HA is

```
begin
  s <= in1 xor in2;
  c <= in1 and in2;
```

```
end Dataflow;
```

testbench :

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
```

```
entity HA_tb is
  -- Port ( );
end HA_tb;
```

architecture tb of HA_tb is

```
  component HA is
    port(
      A : in std_logic;
      B : in std_logic;
      Sum : out std_logic;
      Carry : out std_logic);
  end component;
```

```
  signal a,b,sum,carry: std_logic;
```

```
begin
```

```
  uut: HA
```

```

port map(
  A=>a,
  B=>b,
  Sum=>sum,
  Carry=>carry
);
stim: process
begin

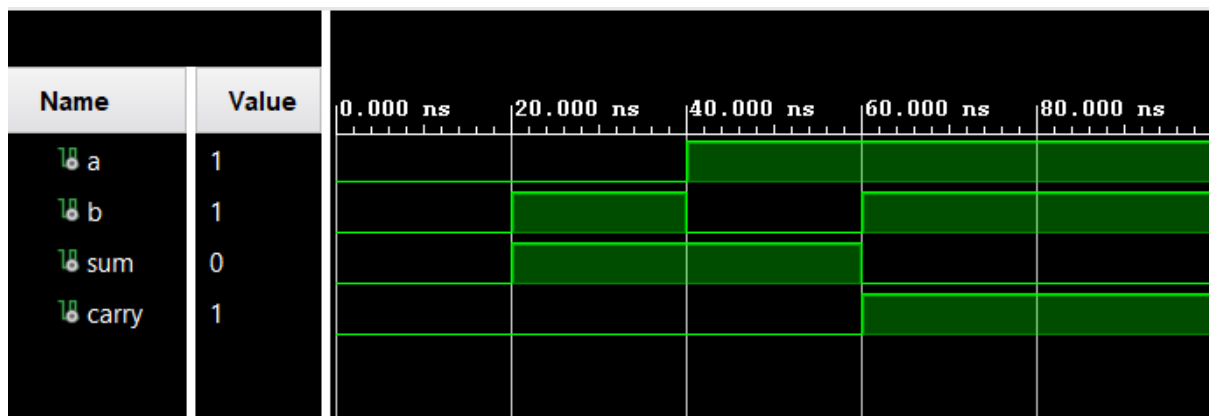
a <= '0';b <= '0';wait for 20 ns;
a <= '0';b <= '1';wait for 20 ns;
a <= '1';b <= '0';wait for 20 ns;
a <= '1';b <= '1';wait for 20 ns;

wait;
end process;

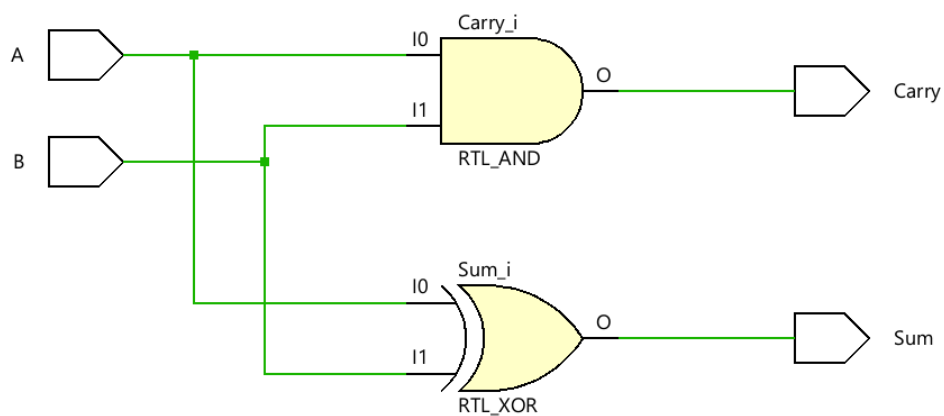
end tb;

```

Και για την προσομοίωση :



RTL schematic :



Για να βρούμε το critical path κάνουμε report timing summary και λαμβάνουμε τα εξής :

Name	Slack ^{^1}	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay
↳ Path 1	∞	3	4	2	B	Sum	5.377	3.778	1.599
↳ Path 2	∞	3	4	2	B	Carry	5.351	3.752	1.599

Καταλαβαίνουμε εύκολα ότι το critical path είναι το path 1, αφού είναι αυτό που παράγει την μεγαλύτερη καθυστέρηση (total delay).

2. Full adder - Structural

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity FA is
  Port ( A : in STD_LOGIC;
        B : in STD_LOGIC;
        Cin : in STD_LOGIC;
        Sum : out STD_LOGIC;
        Cout : out STD_LOGIC);
end FA;

architecture Structural of FA is

  component HA
    port(in1,in2 : in std_logic;
         s,c : out std_logic);
  end component;

  component or1
    port(in1,in2 : in std_logic;
         o1: out std_logic);
  end component;

  signal sum1,co1,co2:std_logic;

begin
  X1 : HA port map(a,b,sum1,co1);
  X2 : HA port map(sum1,Cin,sum,co2);
  X3 : or1 port map(co1,co2,Cout);

end Structural;
```

Και για το testbench :

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
```

```

entity FA_tb is
-- Port ( );
end FA_tb;

architecture Behavioral of FA_tb is
component FA
    port ( A : in STD_LOGIC;
          B : in STD_LOGIC;
          Cin : in STD_LOGIC;
          Sum : out STD_LOGIC;
          Cout : out STD_LOGIC);
end component;

    signal A, B, Cin, Sum, Cout : std_logic;

begin
    uut: FA
    port map (
        A => A,
        B => B,
        Cin => Cin,
        Sum => Sum,
        Cout => Cout
    );

    testing : process
    begin

        A <= '0'; B <= '0'; Cin <= '0'; wait for 50 ns;
        A <= '0'; B <= '0'; Cin <= '1'; wait for 50 ns;
        A <= '0'; B <= '1'; Cin <= '0'; wait for 50 ns;
        A <= '0'; B <= '1'; Cin <= '1'; wait for 50 ns;
        A <= '1'; B <= '0'; Cin <= '0'; wait for 50 ns;
        A <= '1'; B <= '0'; Cin <= '1'; wait for 50 ns;
        A <= '1'; B <= '1'; Cin <= '0'; wait for 50 ns;
        A <= '1'; B <= '1'; Cin <= '1'; wait for 50 ns;

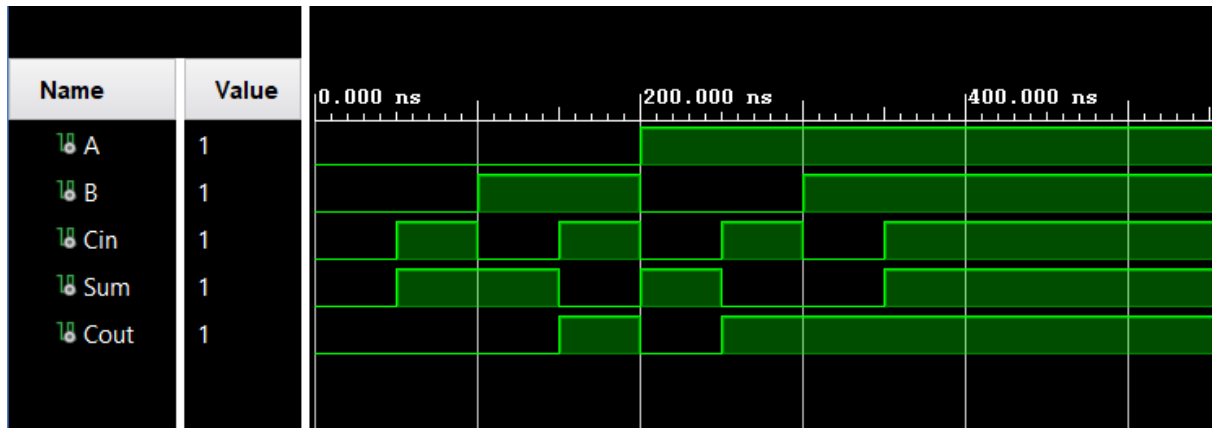
        wait;

    end process;

end Behavioral;

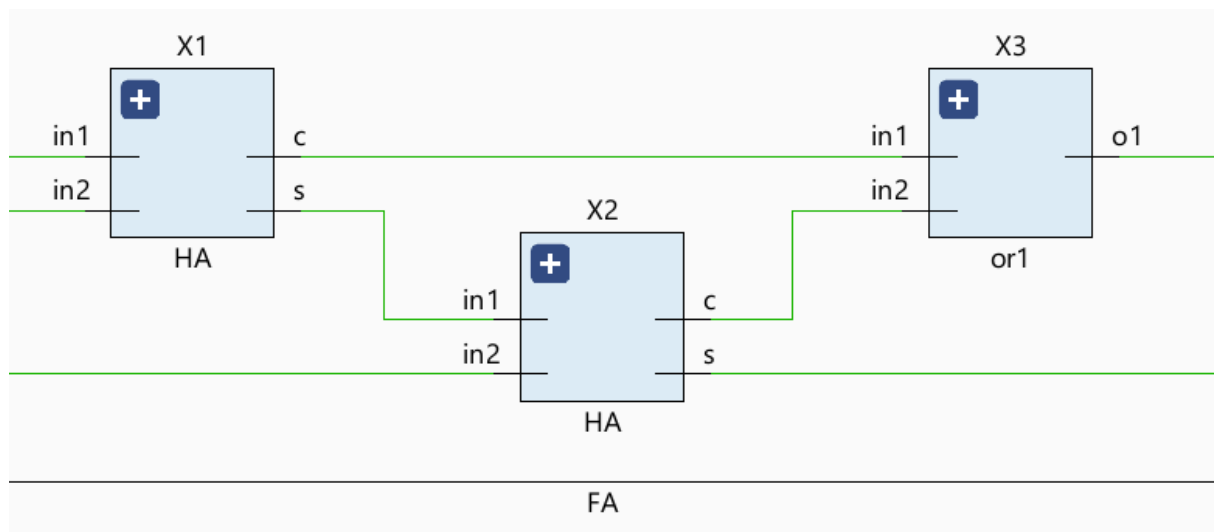
```

Πατήσαμε run simulation και λάβαμε τα εξής :



όπως βλέπουμε από τις γραφικές μας, ο full adder λειτουργεί ορθά.

Και το schematic :



Όπως φαίνεται από το schematic αλλά και από τον κώδικα, υλοποιήσαμε τον full adder χρησιμοποιώντας 2 half adders και μία πύλη or την οποία υλοποιήσαμε ξεχωριστά. Αυτό μας το επιτρέπει η αρχιτεκτονική structural αφού καλούμε τα προηγούμενα sources σαν components στα επόμενα και καθαυτό τον τρόπο δεν χρειάζεται να τα ξαναφτιάχνουμε κάθε φορά

3. 4-bit Parallel Adder - Structural

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity Parallel_FA is
port(
    a : in STD_LOGIC_VECTOR(3 downto 0);
    b : in STD_LOGIC_VECTOR(3 downto 0);
    carryin : in std_logic;
    carryout : out STD_LOGIC;
    sum : out STD_LOGIC_VECTOR(3 downto 0)
);
end Parallel_FA;

architecture Structural of Parallel_FA is

    component FA is
    port( A : in std_logic;
    B: in std_logic;
    Cin : in std_logic;
    Sum : out std_logic;
    Cout : out std_logic);
    end component;

    signal c : std_logic_vector(2 downto 0);

begin

    x4 : FA port map(a(0),b(0),carryin,sum(0),c(0));
    x5 : FA port map(a(1),b(1),c(0),sum(1),c(1));
    x6 : FA port map(a(2),b(2),c(1),sum(2),c(2));
    x7 : FA port map(a(3),b(3),c(2),sum(3),carryout);
end Structural;
```

Testbench :

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity parallel_fa_tb is
-- Port ( );
end parallel_fa_tb;

architecture Behavioral of parallel_fa_tb is
```

```

-- Declare the component for the 4-bit adder
component parallel_fa
  port(
    A, B : in std_logic_vector(3 downto 0);
    Carryin : in std_logic;
    Carryout : out std_logic;
    Sum : out std_logic_vector(3 downto 0)

  );
end component;

-- Declare signals for inputs and outputs
signal A, B : std_logic_vector(3 downto 0);
signal Carryin : std_logic;
signal Sum : std_logic_vector(3 downto 0);
signal Carryout : std_logic;

begin
  uut: parallel_fa
    port map (
      A => A,
      B => B,
      Carryin => Carryin,
      carryout => carryout,
      Sum => Sum
    );

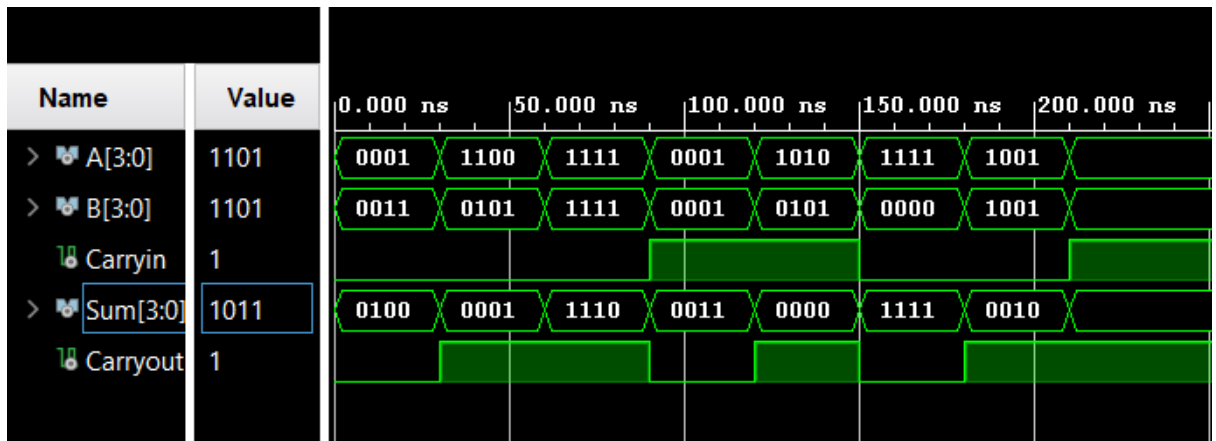
  testing: process
    begin

      A <= "0001"; B <= "0011"; Carryin <= '0'; wait for 30 ns;
      A <= "1100"; B <= "0101"; Carryin <= '0'; wait for 30 ns;
      A <= "1111"; B <= "1111"; Carryin <= '0'; wait for 30 ns;
      A <= "0001"; B <= "0001"; Carryin <= '1'; wait for 30 ns;
      A <= "1010"; B <= "0101"; Carryin <= '1'; wait for 30 ns;
      A <= "1111"; B <= "0000"; Carryin <= '0'; wait for 30 ns;
      A <= "1001"; B <= "1001"; Carryin <= '0'; wait for 30 ns;
      A <= "1101"; B <= "1101"; Carryin <= '1'; wait for 30 ns;

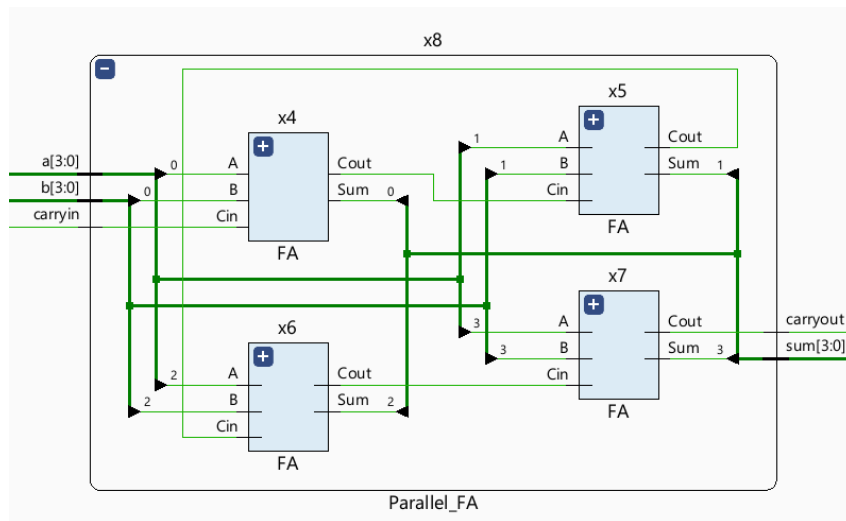
      wait;
    end process;
end Behavioral;

```

Simulation :



Schematic :



4. BCD Full Adder - Structural

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;
```

entity BCD_FA is

```
Port (a : in STD_LOGIC_VECTOR(3 downto 0);
      b : in STD_LOGIC_VECTOR(3 downto 0);
      carryin : in std_logic;
      carryout : out STD_LOGIC;
      sum : out STD_LOGIC_VECTOR(3 downto 0)
    );
```

end BCD_FA;

architecture Structural of BCD_FA is

component Parallel_FA

```
port(
    a : in STD_LOGIC_VECTOR(3 downto 0);
    b : in STD_LOGIC_VECTOR(3 downto 0);
    carryin : in std_logic;
    carryout : out STD_LOGIC;
    sum : out STD_LOGIC_VECTOR(3 downto 0)
);
end component;
```

```
signal sum1 : std_logic_vector(3 downto 0);
signal carry1 : std_logic;
```

```
signal binput : std_logic_vector(3 downto 0);
```

```
signal correction : std_logic;
signal carry2 : std_logic;
begin
    correction <= ((sum1(3) and (sum1(2) or sum1(1))) or carry1);
    binput(0) <= '0';
    binput(1) <= correction;
    binput(2) <= correction;
    binput(3) <= '0';
```

```
x8 : Parallel_FA port map (a,b,carryin,carry1,sum1);
x9 : parallel_fa port map (sum1,binput,'0',carry2,sum);
carryout <= correction;
end Structural;
```

Testbench :

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;
```

```
entity bcd_fa_tb is
-- Port ( );
end bcd_fa_tb;
```

architecture Behavioral of bcd_fa_tb is

```
component BCD_FA
    Port (a : in STD_LOGIC_VECTOR(3 downto 0);
```

```

        b : in STD_LOGIC_VECTOR(3 downto 0);
        carryin : in std_logic;
        carryout : out STD_LOGIC;
        sum : out STD_LOGIC_VECTOR(3 downto 0)
    );
end component;

signal A, B, Sum : std_logic_vector(3 downto 0);
signal Cin, Cout : std_logic;
begin

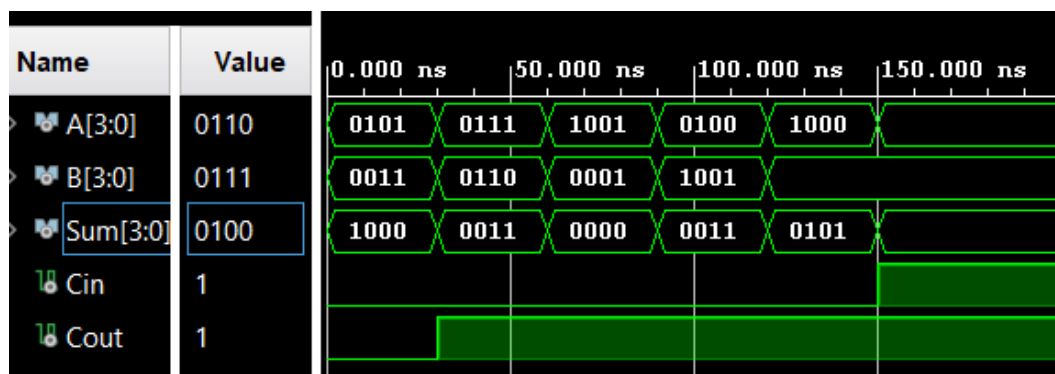
UUT: BCD_FA
    port map (
        A => A,
        B => B,
        Carryin => Cin,
        Sum => Sum,
        Carryout => Cout
    );
process
begin

    A <= "0101";B <= "0011";Cin <= '0'; wait for 30 ns;
    A <= "0111";B <= "0110"; Cin <= '0'; wait for 30 ns;
    A <= "1001";B <= "0001";Cin <= '0';wait for 30 ns;
    A <= "0100";B <= "1001";Cin <= '0';wait for 30 ns;
    A <= "1000";B <= "0111";Cin <= '0';wait for 30 ns;
    A <= "0110";B <= "0111";Cin <= '1';wait for 30 ns;

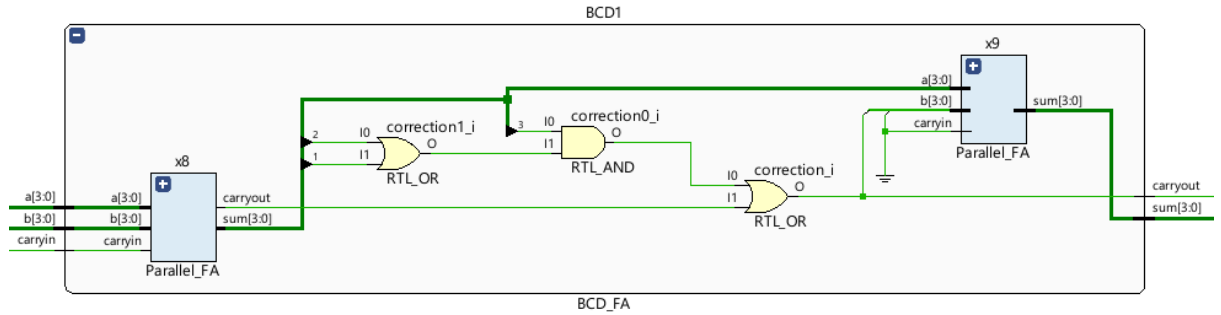
    wait;
end process;
end Behavioral;

```

Simulation :



RTL Schematic :



5. 4-BCD Parallel Adder - Structural

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;
```

```
entity bcd4dig is
  Port (Ain: in std_logic_vector(15 downto 0);
        Bin: in std_logic_vector(15 downto 0);
        cin: in std_logic;
        cout: out std_logic;
        sum: out std_logic_vector(15 downto 0));
end bcd4dig;
```

```
architecture structural of bcd4dig is
  component BCD_FA is
    Port ( A: in STD_LOGIC_VECTOR (3 downto 0);
          B : in STD_LOGIC_VECTOR (3 downto 0);
          Carryin : in STD_LOGIC;
          Carryout : out std_logic;
          sum : out STD_LOGIC_VECTOR (3 downto 0));
  end component;
```

```
signal coutp_int: std_logic_vector(3 downto 0);
```

```
signal A1, A2, A3, A4: std_logic_vector(3 downto 0);
signal B1, B2, B3, B4: std_logic_vector(3 downto 0);
```

```
signal s1, s2, s3, s4: std_logic_vector(3 downto 0);
```

begin

```
A1 <= Ain(3 downto 0);
A2 <= Ain(7 downto 4);
```

```
A3 <= Ain(11 downto 8);
A4 <= Ain(15 downto 12);
```

```
B1 <= Bin(3 downto 0);
B2 <= Bin(7 downto 4);
B3 <= Bin(11 downto 8);
B4 <= Bin(15 downto 12);
```

```
BCD1: BCD_FA port map (A1,B1,Cin,coutp_int(0),s1);
BCD2: BCD_FA port map (A2,B2,coutp_int(0),coutp_int(1),s2);
BCD3: BCD_FA port map (A3,B3,coutp_int(1),coutp_int(2),s3);
BCD4: BCD_FA port map (A4,B4,coutp_int(2),cout,s4);
```

```
sum <= s4 & s3 & s2 & s1;
```

```
end structural;
```

Testbench :

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;
```

```
entity bcd4dig_tb1 is
-- Port ( );
end bcd4dig_tb1;
```

```
architecture Behavioral of bcd4dig_tb1 is
```

```
component bcd4dig
    Port (Ain: in std_logic_vector(15 downto 0);
          Bin: in std_logic_vector(15 downto 0);
          cin: in std_logic;
          cout:out std_logic;
          sum:out std_logic_vector(15 downto 0));
end component;
```

```
    signal Ain, Bin, sum : std_logic_vector(15 downto 0);
    signal cin, cout      : std_logic;
```

```
begin
```

```
    DUT: bcd4dig port map(
        Ain  => Ain,
        Bin  => Bin,
        cin  => cin,
        cout => cout,
        sum  => sum
    );
```

```

process
begin

    Ain <= "0000000000000000";
    Bin <= "0000000000000000";
    cin <= '0';
    wait for 30 ns;

    Ain <= "0000000000000001";
    Bin <= "0000000000000001";
    cin <= '0';
    wait for 30 ns;

    Ain <= "0000000000000110";
    Bin <= "000000000000011";
    cin <= '0';
    wait for 30 ns;

    Ain <= "0000000000000101"; --5+5
    Bin <= "0000000000000101"; --output ends in 0001 0000
    cin <= '0';
    wait for 30 ns;

    Ain <= "0000000000001010";
    Bin <= "000000000000101";
    cin <= '0';
    wait for 30 ns;

    Ain <= "0000000000001001"; --9
    Bin <= "0000000000001001"; --9
    cin <= '0';
    wait for 30 ns;

    Ain <= "0001001000110100"; -- 1234
    Bin <= "0100001100100001"; -- 4321
    cin <= '0';
    wait for 30 ns;

    Ain <= "1001100110011001"; -- 9999
    Bin <= "0000000000000001"; -- 0001
    cin <= '0';
    wait for 30 ns;

    Ain <= "0101011001111000"; -- 5678
    Bin <= "0001001000110100"; -- 1234
    cin <= '0';
    wait for 30 ns;
    wait;

```

```

end process;
end Behavioral;

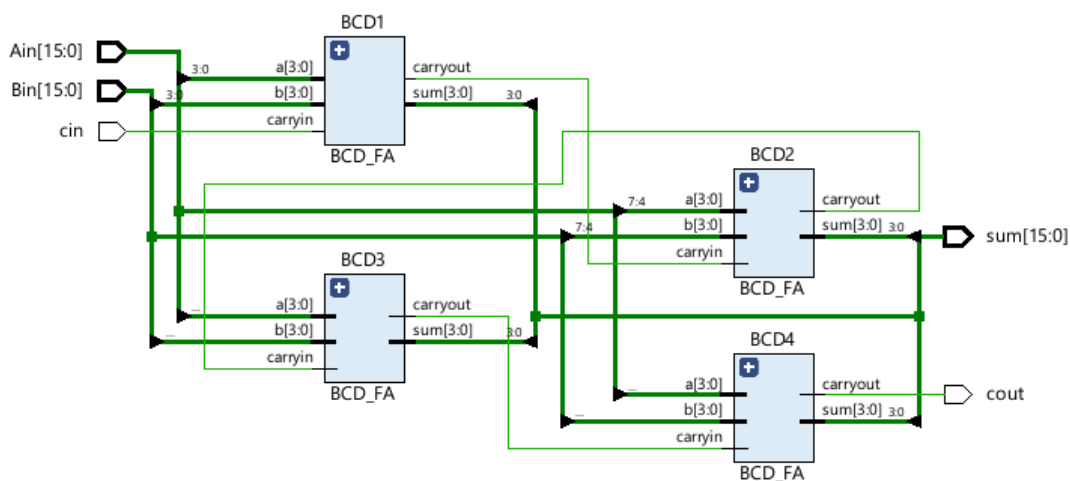
```

Simulation :

Name	Value	0.000 ns	10.000 ns	20.000 ns	30.000 ns	40.000 ns	50.000 ns	60.000 ns	70.000 ns	80.000 ns	90.000 ns	100.000 ns	110.000 ns	120.000 ns	130.000 ns
Ain[15:0]	00000000	0000000000000000	0000000000000000	0000000000000000	0000000000000001	0000000000000001	0000000000000010	0000000000000011	0000000000000011	0000000000000011	0000000000000011	0000000000000011	0000000000000011	0000000000000011	0000000000000011
Bin[15:0]	00000000	0000000000000000	0000000000000000	0000000000000000	0000000000000001	0000000000000001	0000000000000010	0000000000000011	0000000000000011	0000000000000011	0000000000000011	0000000000000011	0000000000000011	0000000000000011	0000000000000011
sum[15:0]	00000000	0000000000000000	0000000000000000	0000000000000000	0000000000000001	0000000000000001	0000000000000010	0000000000000011	0000000000000011	0000000000000011	0000000000000011	0000000000000011	0000000000000011	0000000000000011	0000000000000011
cin	0														
cout	0														

Name	Value	150.000 ns	160.000 ns	170.000 ns	180.000 ns	190.000 ns	200.000 ns	210.000 ns	220.000 ns	230.000 ns	240.000 ns	250.000 ns	260.000 ns	270.000 ns
Ain[15:0]	01010110	0000000000000001	0000000000000001	0000000000000001	0001001000110100	0001001000110100	0001001000110100	0001001000110100	0001001000110100	0001001000110100	0001001000110100	0001001000110100	0001001000110100	0001001000110100
Bin[15:0]	00010010	0000000000000001	0000000000000001	0000000000000001	0100001100100001	0100001100100001	0100001100100001	0100001100100001	0100001100100001	0100001100100001	0100001100100001	0100001100100001	0100001100100001	0100001100100001
sum[15:0]	01101001	0000000000000001	0000000000000001	0000000000000001	0101010101010101	0101010101010101	0101010101010101	0101010101010101	0101010101010101	0101010101010101	0101010101010101	0101010101010101	0101010101010101	0101010101010101
cin	0													
cout	0													

Schematic :



Και το critical path :

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock
Path 1	∞	10	11	7	Bin[0]	sum[14]	9.549	4.648	4.901	∞	input port clock
Path 2	∞	10	11	7	Bin[0]	cout	9.543	4.642	4.901	∞	input port clock
Path 3	∞	10	11	7	Bin[0]	sum[13]	9.543	4.642	4.901	∞	input port clock
Path 4	∞	10	11	7	Bin[0]	sum[15]	9.543	4.642	4.901	∞	input port clock
Path 5	∞	9	10	7	Bin[0]	sum[10]	8.952	4.524	4.428	∞	input port clock
Path 6	∞	9	10	7	Bin[0]	sum[12]	8.952	4.524	4.428	∞	input port clock
Path 7	∞	9	10	7	Bin[0]	sum[9]	8.952	4.524	4.428	∞	input port clock
Path 8	∞	8	9	7	Bin[0]	sum[11]	8.345	4.394	3.951	∞	input port clock
Path 9	∞	7	8	7	Bin[0]	sum[5]	7.767	4.276	3.491	∞	input port clock
Path 10	∞	7	8	7	Bin[0]	sum[6]	7.767	4.276	3.491	∞	input port clock

Το critical path είναι το path1 αφού είναι αυτο με το μεγαλύτερο total Delay